Biochemistry Cume Exam 1

1. **(30 points).** Draw the complete structure(s) of the oligopeptides below at pH=7.

a) A-H

b) D-P-C

c) Show how an oligopeptide of Leu and Lys can be either a <u>branched-chain</u> or a <u>straight-chain</u> structure.

2. **(30 points).** A mixture of the following amino acids is subjected to paper electrophoresis @ pH=3.9: A, S, F, L, R, D, H. Which of these amino acids will migrate toward the anode? Which will migrate toward the cathode?

3. **(30 points).** a) Assume you have a solution containing 0.1 mol of Ala adjusted to pH=0.5 with HCl. You begin adding 1.0M NaOH to the solution. Sketch the resulting titration curve showing all inflection points.

b) Sketch a similar titration curve for a solution containing 0.1 mol of His at pH=0.5 by titrating 1.0M NaOH to the solution.

4. **(10 points).** Indicate the relative mobilities of amino acids in the following mixtures during paper chromatography in a butanol:acetic acid:water system. Assume the pH=4.5 for the mixture in the aqueous phase.

a) Isoleucine, Lysine

b) Phenylalanine, Serine